Detecting Quarry Pond Remnants on a Japanese Island Heritage Site Using Sentinel-2 Imagery and Open-Source Remote Sensing Tools

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Open-source Sentinel-2 processing revealed 113 spring and 145 summer intra-island water polygons on Kitagi Island; their distribution is consistent with historical quarrying patterns, while individual quarry-pond identities still require field validation.

1 · Background and heritage

Kitagi Island (Kitagi-shima), in Kasaoka City, Okayama Prefecture, is a granite-quarrying island in Japan's Seto Inland Sea. Quarrying here extends back to the early seventeenth century.

127 active quarry sites (dojo) recorded at the 1957 peak

As the industry declined, abandoned excavations filled with rainwater and groundwater, forming isolated ponds enclosed by steep granite walls. In 2019 the island's stone culture became part of Japan's national heritage under the "Stone Islands of Setouchi" program. No systematic spatial inventory of these quarry pond remnants has been published.

2 · Research question

Can open satellite imagery and reproducible open-source tools map water bodies associated with former quarry sites, and do their spatial patterns correspond to the island's quarrying history?

(a) Seto Inland Sea, western Japan

(b) Kasaoka Islands, Okayama Pref.

Study area: Kitagi Island, Kasaoka City, Okayama Prefecture. Basemap: GSI Tiles (English), Geospatial Information Authority of Japan.

3 · Data and study area

- Sentinel-2 L2A imagery accessed through the Microsoft Planetary Computer STAC API
- Spring scene 2025-03-23 (cloud cover 0.0%); summer scene 2025-08-02 (cloud cover 0.7%)
- 10 m analysis grid: B02/B03/B04/B08 are native 10 m; B11 (SWIR) is resampled from its native 20 m. Mixed pixels limit very small water bodies
- Minimum reported polygon area: 100 m²

Water-filled quarry remnants ("Choba lake") enclosed by granite walls, Kitagi Island. Photos: Noboru Otsuka.

4 · Method — open-source pipeline

Sentinel-2 L2A; 10 m analysis grid; B11 SWIR resampled from 20 m

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STAC search and band access — Microsoft Planetary Computer

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Water index union: NDWI + MNDWI

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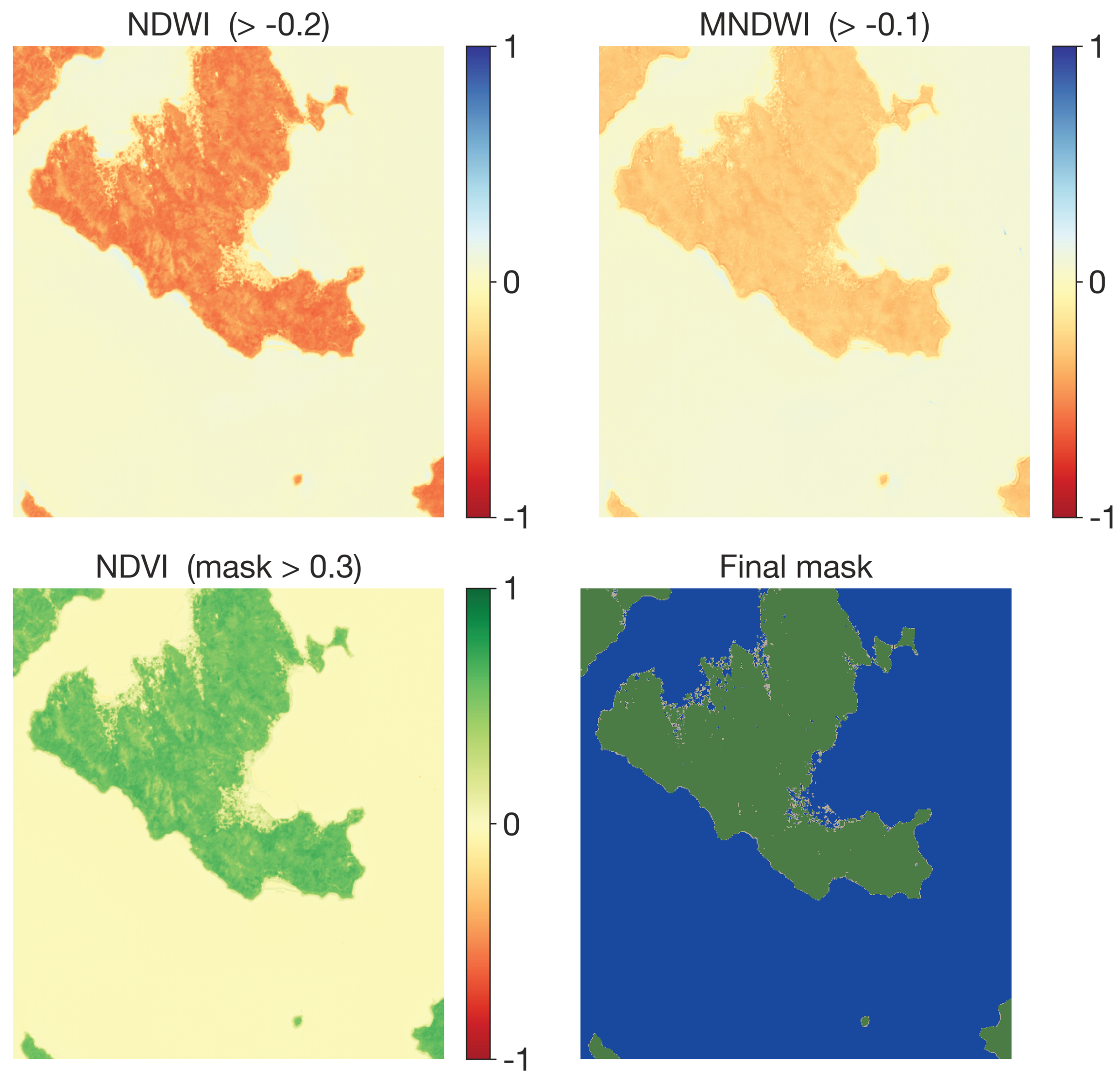
NDVI vegetation mask, then water polygon extraction (≥100 m²)

NDWI = (Green − NIR) / (Green + NIR)
MNDWI = (Green − SWIR) / (Green + SWIR)
NDVI = (NIR − Red) / (NIR + Red)

(NDWI > −0.2 OR MNDWI > −0.1) AND NOT (NDVI > 0.3)

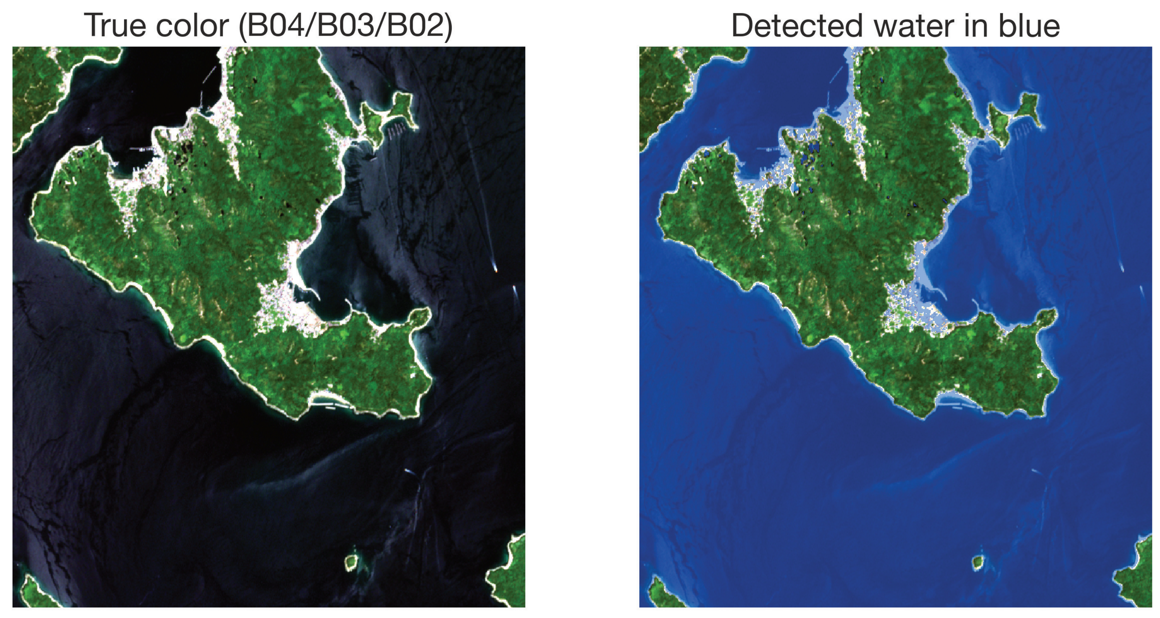
The union condition favors sensitivity to small water bodies affected by spectral mixing at 10 m resolution, while the NDVI mask excludes vegetation-like pixels.



NDWI, MNDWI, NDVI and the final mask (blue = water, green = vegetation, gray = other), Sentinel-2 L2A, 2025-08-02. B11 (SWIR, native 20 m) resampled to the 10 m grid. Contains modified Copernicus Sentinel data [2025].

10 m Sentinel-2 spatial resolution — the key detection limit

The summer NDVI mask excluded only 9 pixels, indicating limited overlap between the detected water candidates and vegetation zones in this granite-dominated landscape.



Detected water candidates highlighted in blue; not individual quarry validation. Contains modified Copernicus Sentinel data [2025].

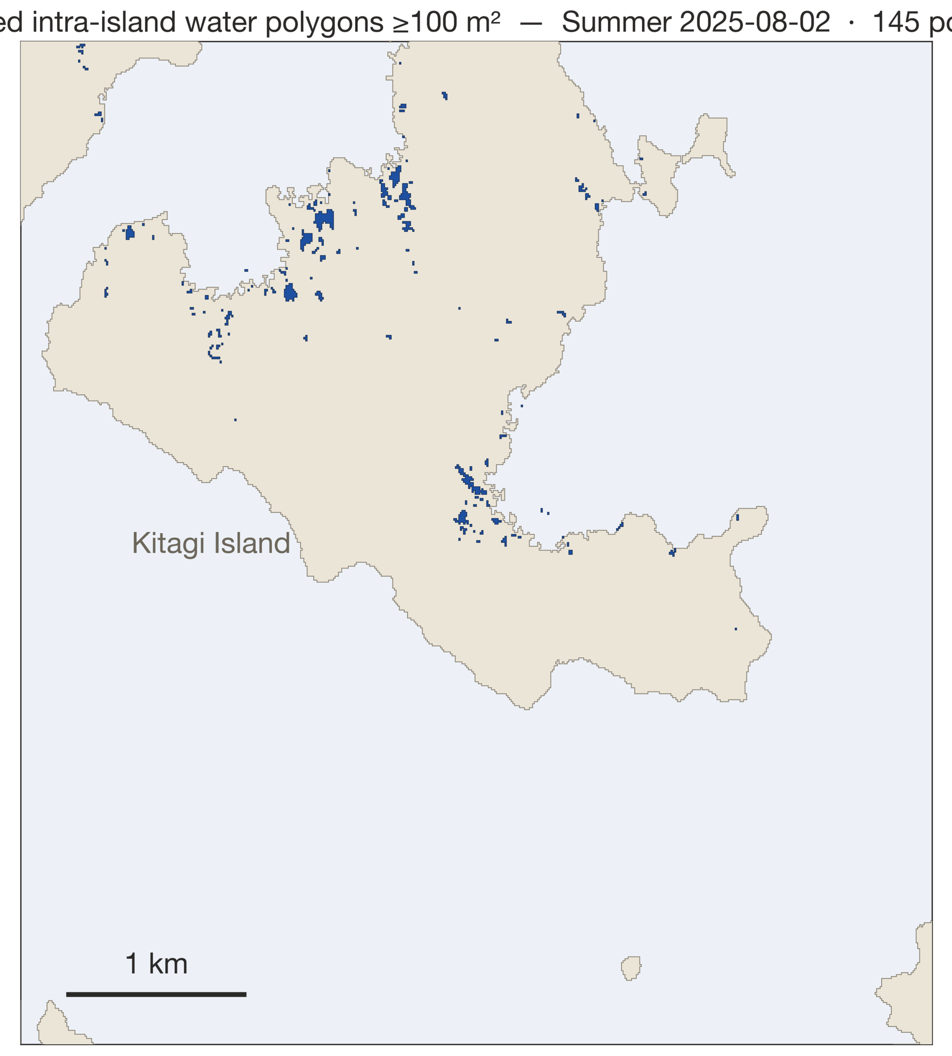
5 · Results — seasonal detection

Detected intra-island water polygons ≥100 m²

113	145
Spring · 2025-03-23	Summer · 2025-08-02
water polygons ≥100 m²	water polygons ≥100 m²

Spring count: 113 reported polygons; the exact historical run configuration is not preserved in the current repository.

Detected intra-island water polygons ≥100 m² — Summer 2025-08-02 · 145 polygons



Summer result: 145 detected intra-island water polygons (blue) on a land/sea outline derived from the same Sentinel-2 scene. Contains modified Copernicus Sentinel data [2025].

Summer imagery detected 145 intra-island water polygons. The detections were concentrated in northern, southeastern, central, and western parts of the island, a pattern consistent with historical quarrying records.

These are detected water polygons, not individually field-confirmed quarry ponds. Natural ponds, reservoirs, shadows, and other false positives may remain.

6 · Interpretation and limitations

- The count and distribution of detected polygons are consistent with the historical quarrying context; the 145-to-127 comparison is a scale comparison, not a one-to-one match.
- Sentinel-2's 10 m resolution causes spectral mixing for narrow or small ponds.
- Negative thresholds improve sensitivity but can add false positives from dark rock or shadows.
- Individual quarry-pond identity and accuracy metrics require field validation.
- Season changes which water bodies are most detectable; spring and summer results are not interchangeable observations.

Conclusion and reuse

A lightweight, reproducible workflow can provide a first spatial inventory of quarry-pond candidates and a practical base layer for field validation and heritage documentation. Results are exported as GeoJSON and GeoTIFF. The workflow uses open-source Python libraries — rasterio, numpy, shapely, pystac-client, planetary-computer, and folium — and could be extended to other quarried islands in the Seto Inland Sea.

References

McFeeters, S. K. (1996) The use of the Normalized Difference Water Index (NDWI) in the delineation of open water features. Int. J. Remote Sensing, 17(7), 1425–1432.
Xu, H. (2006) Modification of normalised difference water index (NDWI) to enhance open water features in remotely sensed imagery. Int. J. Remote Sensing, 27(14), 3025–3033.
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Japan Heritage story "Islands of stone": stone-islands.jp/en/story/

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